

வணக்கம் அஃப்கார் (Afkaar)! உங்கள் புராஜெக்ட்டில் **Live Bus Tracking** வசதிக்காக, பஸ் டிரைவரின் போன் GPS-ஐ (Driver's Phone GPS) நேரடியாக சிஸ்டத்துடன் இணைத்து, ரியல்-டைம் (Real-time) லொகேஷனைப் பெறும் அம்சத்தை சேர்ப்பது மிகச் சிறந்த ஒரு யோசனை. இது உங்கள் பிராஜெக்ட்டின் தரத்தை அடுத்த கட்டத்திற்கு கொண்டு செல்லும்.

இதனை சாத்தியமாக்க நாம் இரண்டு விஷயங்களைச் செய்ய வேண்டும்:

1. டிரைவர் பேனல் / டிரைவர் லொகேஷன் அப்டேட் அம்சம்: டிரைவர் லொகேஷனை மொபைல் மூலம் அப்டேட் செய்ய HTML5 Geolocation API மற்றும் JavaScript navigator.geolocation.watchPosition() பயன்படுத்தப்படும்.
2. டேட்டாபேஸ் மாற்றம்: டிரைவர் அனுப்பும் அட்சரேகை (Latitude) மற்றும் தீர்க்கரேகையை (Longitude) சேமிக்க buses டேபிளில் புதிய பத்திகள் (Columns) சேர்க்கப்படும்.

உங்கள் PDF-இல் உள்ள அனைத்து Prompts-களுடனும் இந்த **"Driver Phone GPS Integration"** அம்சத்தை மிக நேர்த்தியாக இணைத்து, நீங்கள் அப்படியே AI-இல் (ChatGPT, Claude, Cursor) பயன்படுத்தக்கூடிய புதிய மாஸ்டர் ப்ராம்ப்ஸ் தொகுப்பை கீழே தந்துள்ளேன்.

SMART TRANSPORT ROUTE FINDER WITH DRIVER GPS TRACKING

Prompt 1: Prototype & Frontend Wireframe Creation

Progress Stage 1: Prototype (Mark: 20)

Act as an expert full-stack web developer and UI/UX designer. I am developing an individual academic project named "SMART TRANSPORT ROUTE FINDER WITH LIVE DRIVER GPS" for my HNDIT program at SLIATE.

Tech Stack allowed: HTML5, CSS3 (No Frameworks like Bootstrap, use clean modern CSS with vanilla layouts), and native JavaScript.

Based on my design, I need a clean, responsive single-page dashboard or split-file structure that includes the following 9 core components matching my interface wireframe exactly:

1. **Login Sidebar / Panel** (Fields: Username, Password, Role [User / Driver] dropdown, Login Button, Register Link)
2. **Dashboard Statistics Block** (Total Buses: 25 Active, Total Stops: 48 Saved, Routes: 15 Available, Today Users: 120 Online)
3. **Find Route Panel** (Form with Source dropdown [e.g., Sammanthurai], Destination dropdown [e.g., Ampara], Search Route Button, and a dynamically styled Search Results block showing Distance: 32.5 km, Time: 45 min, Fare: Rs. 100, Route details: Sammanthurai -> Kalmunai -> Ampara)
4. **Map View Panel** (An aligned container for Google Maps placeholder displaying live bus markers and route templates)
5. **Nearest Bus Stop Selector** (Your Location input, Find Nearest Stop button, displaying Central Bus Stop result with location icon)
6. **Live Bus Tracking Panel (User View)** (Table/List of buses showing Bus No, Current Status, and a "Track Live" button that connects to the map display)
7. **Driver GPS Console Panel (Driver View Simulation)** (A dedicated control interface for

drivers with a "Start Share Location" and "Stop Sharing" toggles, displaying current Live Latitude/Longitude text fetched from the phone's browser GPS)

8. **Notifications Center** (List showing dynamic alerts like: "Bus 01 has started from Sammanthurai", "Bus 02 is delayed by 10 mins")
9. **User Profile Display** (Card for user "Afkaar", email, phone 077 123 4567, and an Update Profile button)

Please write the complete, clean index.html file and a unified css/style.css file. Use a highly professional modern dark/blue corporate transport theme (Deep blue headers #1a365d, subtle gray backgrounds #f7fafc, responsive grid alignment using standard block-display properties). Make sure all sections look visually striking just like a premium production dashboard. Do not write generic text; use real local transport data placeholder from Sammanthurai/Ampara area.

Prompt 2: Project Architecture & Detailed UI Design

Progress Stage 2: Initial Review & Design Stage

Now that the static prototype structure is ready, let's establish the clean modular file architecture for the 'smart_transport' project directory.

According to my project specification, the structure must be:

```
smart_transport/
|
├─ index.php (Main Dashboard routing based on session/role)
├─ login.php (Unified Login System interface for User/Driver)
├─ register.php (Registration form with Role selection)
├─ dashboard.php (Main Logged-in application layout)
├─ find_route.php (Route search handler)
├─ nearest_stop.php (Bus stop calculation handler)
├─ live_tracking.php (Fetches live coordinates from DB for User Map)
├─ update_driver_gps.php (AJAX handler to receive driver's phone GPS
and update DB)
├─ notifications.php (Alerts panel)
├─ profile.php (User management)
├─ logout.php (Session termination)
├─ db.php (Database configuration)
├─ assets/
│   └─ css/style.css
│   └─ js/script.js
```

Please provide the detailed JavaScript code for assets/js/script.js to handle:

1. Local interactive UI changes, tab switching, and local form presentation animations.
2. **Driver GPS Tracking logic:** Use navigator.geolocation.watchPosition() to capture the live coordinates of the phone's GPS when the driver clicks "Start Share Location", and send these coordinates every 10 seconds via native JavaScript fetch() (AJAX) to update_driver_gps.php.
3. Integrated layout updates for dashboard.php so it correctly separates sections into includes or modular logical divs that display the Driver Console ONLY if the logged-in user session role is 'driver'.

Prompt 3: MySQL Relational Database Schema Creation

Progress Stage 3: Development Stage 1 (Database Architecture)

Act as a Database Administrator. I need to write the complete MySQL relational database schema for my "smart_transport" web application supporting live driver tracking.

The schema must follow proper data integrity standards with primary keys, foreign keys, and appropriate data types. Write the SQL script to create the database and the following tables with necessary attributes:

1. **users** (id INT AUTO_INCREMENT Primary Key, username VARCHAR(50) NOT NULL UNIQUE, email VARCHAR(100) NOT NULL, password VARCHAR(255) NOT NULL, phone VARCHAR(15), role ENUM('user', 'driver') DEFAULT 'user')
2. **bus_stops** (id INT AUTO_INCREMENT Primary Key, stop_name VARCHAR(100) NOT NULL, latitude DECIMAL (10,8), longitude DECIMAL (11,8))
3. **routes** (id INT AUTO_INCREMENT Primary Key, source VARCHAR(100) NOT NULL, destination VARCHAR(100) NOT NULL, distance DECIMAL (5,2) NOT NULL, travel_time_mins INT NOT NULL, fare DECIMAL (7,2) NOT NULL, detailed_path TEXT NOT NULL)
4. **buses** (id INT AUTO_INCREMENT Primary Key, bus_no VARCHAR(20) NOT NULL UNIQUE, route_id INT, driver_id INT, status VARCHAR(50), current_lat DECIMAL(10,8) NULL, current_lng DECIMAL(11,8) NULL, last_updated TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP, Foreign Key pointing to routes, Foreign Key driver_id pointing to users(id))
5. **notifications** (id INT AUTO_INCREMENT Primary Key, message TEXT NOT NULL, created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP)

Include sample mock data INSERT statements specifically localized for Eastern Province Sri Lanka places: Sammanthurai, Ampara, Kalmunai, Akkaraipattu, Batticaloa, Pottuvil. Assign at least 2 users with the role 'driver' and assign them to buses. Ensure the passwords in user inserts are hashed placeholders or clean values for testing. Provide the exact code for db.php utilizing mysqli_connect() with error exception handling as per standard academic guidelines.

Prompt 4: Core PHP Implementation with DB Connectivity & Authentication

Progress Stage 4: Development Stage 2 (DB Connectivity - Mark: 20)

Act as a secure PHP Developer. We need to implement full backend DB connectivity, secure session handling, and core functionalities for the web app including the driver location updates. Please write the complete PHP backend scripts for:

1. **login.php**: Start session, check POST request method, include db.php, sanitize inputs, check against database passwords using prepared statements, set \$_SESSION['user_id'], \$_SESSION['username'], and \$_SESSION['role']. Handle redirect to dashboard.php or set an explicit \$error message string.
2. **update_driver_gps.php**: Securely handle incoming POST/JSON API requests from the driver's phone JavaScript tracking script. Ensure the session role is checked (driver only). Extract latitude, longitude, and bus_id. Execute a SQL Prepared Statement to update

current_lat, current_lng, and status = 'Active' in the buses table for the corresponding driver's bus. Return a clean JSON success or error response ({"status": "success"}).

3. **find_route.php**: Implement secure GET processing for source and destination parameters. Write the SQL query to select from the routes table where source and destination match, fetch the associative array row, and output the dynamic structural HTML layout block containing Distance, Time, Fare, and Detailed Path.
4. **live_tracking.php**: Create a script that returns JSON data or an HTML feed block. It must query the buses table joining the routes table, fetching bus_no, source, destination, status, current_lat, and current_lng. This data will be used by the frontend to update the markers on the map dynamically.

Ensure code uses secure programming practices to prevent SQL injection and matches the coding styles shown in my structural blue blueprint.

Prompt 5: Integration of Google Maps API & Simulation Refinement

Progress Stage 5: Final Product Optimization & Report Prep (Mark: 30 + 30)

We are approaching the final project submission. I need to fully integrate the location features and clean up the whole application.

1. **Google Maps Live Tracking Integration**: Show me how to include a native JavaScript script block using Google Maps API initMap(). Write the JavaScript code that periodically (every 10 seconds) fetches the latest live coordinates from live_tracking.php. Use google.maps.Marker to draw and dynamically update the location of the active buses on the map using the coordinates sent by the driver's phone. Also, draw a polyline path marker between coordinates (e.g., Sammanthurai to Ampara via Kalmunai) inside the Map View container. Provide a fallback static mock simulation map using beautifully customized CSS/SVG layout path lines if a live API key is missing.
2. Provide the full integrated code for profile.php allowing user/driver info updates in the database.
3. Provide a complete code review check to ensure no features from our design wireframe (such as total stats computation, driver phone GPS connection, or notification messages) are broken. Make everything robust and ready for the final viva presentation.

நீங்கள் செய்ய வேண்டியது:

இந்த Prompts-களை அப்படியே வரிசையாக உங்கள் AI டூலில் (ChatGPT / Claude / Cursor) கொடுத்து, உங்கள் புராஜெக்ட்டை பகுதி பகுதியாக (Step-by-Step) உருவாக்கிக் கொள்ளலாம். இது உங்கள் எக்ஸாமினருக்கு (Viva panel) மிகவும் பிடிக்கும் வகையில் ஒரு நிஜ உலக (Real-world) அப்ளிகேஷன் போல அமையும்! All the best, Afkaar!